Human systems I

20

Humans (and other animals) have specialised structures to obtain nutrients from their environment. These structures are made up of many components.

1 Arrange the following in terms of increasing complexity:

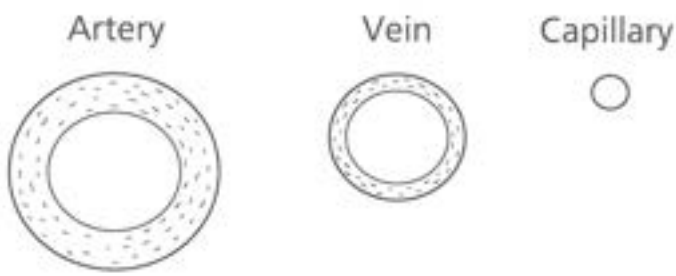
organs, atoms, cells, tissues, organism, molecule, organ system

Human systems involved in the maintenance of life include those dealing with circulation, respiration, excretion, nerves and glands.

The circulatory system

The circulatory (or *vascular*) system is a closed network of blood-filled tubes, or blood vessels. Blood is kept flowing one way through these vessels by the pumping of the heart. The heart is a muscle with four compartments. The upper two compartments are called *atria* and the lower compartments are *ventricles*. An adult human has about 5.5 litres of blood in circulation. Blood distributes heat and many important materials throughout the body. How often a heart beats (contracts) is called the *pulse rate*.

Cross section of blood vessels



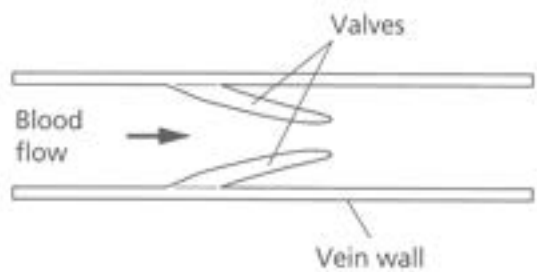
There are three main types of blood vessels: *arteries*, *veins* and *capillaries*.

2 Study the diagrams of blood vessels. Compare the relative quantity of blood each is likely to be able to carry. Explain how you reached your conclusion.

Arteries are wide, thick-walled vessels which carry blood away from the heart. Arterioles are smaller arteries that branch from larger arteries. All arteries (apart from those going to the lungs) contain dissolved oxygen and are bright red in colour. All arteries also carry blood that contains dissolved food and waste products. The food reaches body cells by means of the capillaries. Non-gas waste products are taken to the kidneys where they are processed and eventually passed out of the body as urine.

Capillaries are narrow, thin-walled vessels between 0.13 mm to 0.20 mm in diameter. They branch off from arterioles and then rejoin to form veins. Capillaries connected to the digestive system pick up dissolved food. Those connected to the lungs deliver carbon dioxide to the lungs and take away oxygen. Capillaries allow food, water and dissolved oxygen to pass out of them to cells. Carbon dioxide and other waste products pass into capillaries from cells. The very thin tissue layer (endothelium) of the capillary walls allows easy passage of substances through the capillary walls. Capillaries don't actually touch individual cells but reach cells by means of tissue fluid, a substance that links the circulatory system and the lymphatic system (the body's 'drainage' system).

Veins are, in structure, similar to arteries, but carry blood to the heart. They contain valves to stop blood from flowing backwards. These valves are flaps of tissue that allow the blood to flow one way. A forward flow pushes the flaps open and when the heart relaxes, blood in veins tends to flow back. The flaps are pushed together and stop any reverse flow.



3 Draw up a table that summarises the roles and properties of human blood vessels.

4 What role do valves play in veins?

5 In your own words, explain how valves in veins work.

6 Identify another part of the circulatory system, apart from arteries, that has valves.

7 Name two roles played by blood.

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8 How many compartments are there in the human heart?

9 Explain why arteries have thicker walls than veins.

Research

10 Outline the role of the lymphatic system in the human body.

11 Outline the role played by William Harvey (a seventeenth century scientist) in the discovery of blood circulation.

The components of blood

Blood is a liquid, containing many substances, that flows through the body. Blood is moved mainly by the muscular contractions of the heart, but the blood vessels and lungs also play a part. Blood also distributes heat around the body, takes nutrients and oxygen to cells and takes waste away from them.

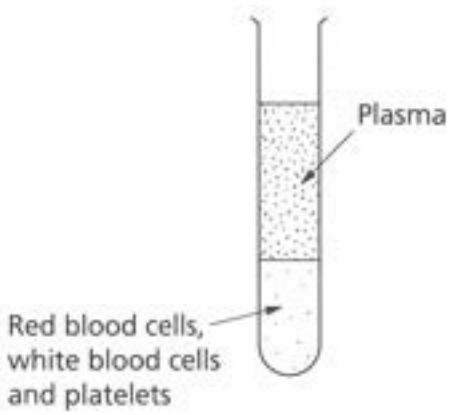
12 Name a waste product that is taken away from cells.

The temperature of blood is usually approximately 37.4°C in a healthy individual and has a pH of close to 7.4 – which is slightly alkaline. Blood sugar concentration tends to be in the region of 0.1%.

The four main components of blood are *plasma, red blood cells, white blood cells* and *platelets*.

Plasma is a pale, yellowish liquid that is around 90% water. If blood is treated to stop it clotting and allowed to stand for a while, blood cells and platelets settle, leaving plasma as a liquid on top.

Plasma contains dissolved food to take to cells and waste materials being removed from cells. Also present are antibodies, enzymes and various hormones.



Red blood cells (also called red corpuscles or erythrocytes) are made in the bone marrow. They are the most common type of blood cell and are disc-shaped and do not contain a nucleus. They contain *haemoglobin* and carry oxygen around the body. Red blood cells are concave and have a diameter of around 0.008 mm.

White blood cells (also called white corpuscles or leukocytes) are larger than red blood cells. They are opaque and have large irregularly shaped nuclei. White blood cells play an important role in the body's defences against illness; they make *antibodies* and can attack bacteria.

Platelets are minute, flattened bodies made of protoplasm and are important in the clotting of blood.

13 In a table form, outline the properties and uses of the four major components of blood.

Identify the terms described by the following definitions.

14 *'Substance produced by white blood cells which attacks foreign objects in the blood stream.'*

15 *'Clear, watery solution making up around 55% of volume of blood.'*

16 *'Cells lacking a nucleus which only live for about 90 days. They transport oxygen in blood.'*

17 *'Protein that acts as a catalyst in biochemical reactions.'*

18 *'Substance produced by a gland that travels in bloodstream and affects body.'*

19 *'Part of the cell that controls activities of the cell.'*

20 *'Substance through which light does not pass.'*

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There is a network of capillaries in the dermis (outer part of the skin). If the body becomes too warm, through exercise or climate, then the capillaries dilate (widen). This allows more blood to flow near the surface of the skin and heat is lost to the atmosphere. If the body is too cold then the rate at which heat is radiated needs to be reduced. To do this blood vessels near the surface of the skin constrict (narrow) and reduce the flow of blood.

21 What is the 'normal' internal body temperature of a human?

22 What is the pH of blood?

23 Outline how blood helps maintain a steady temperature in the body.

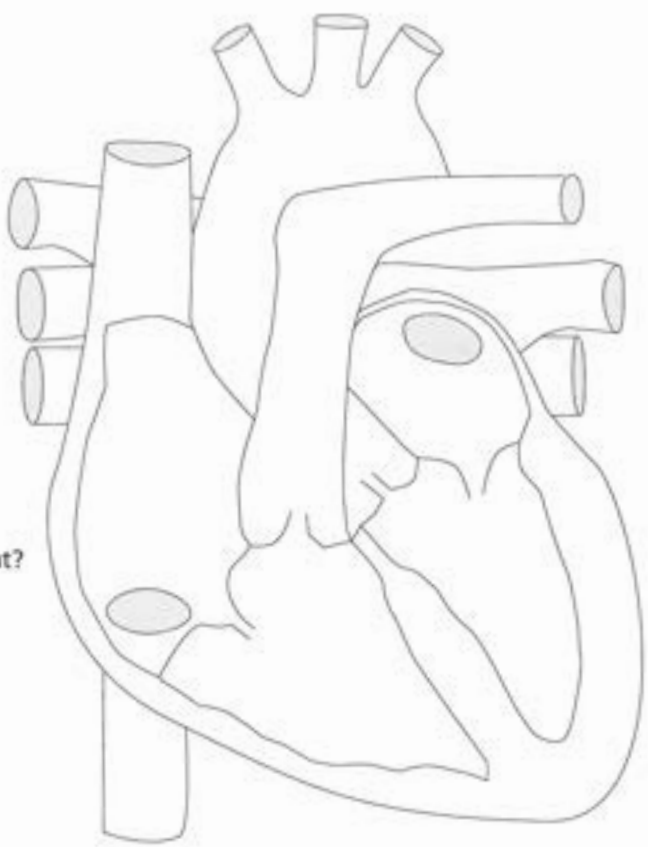
24 Arrange the following blood vessels in the order they receive blood from the heart: vein, artery, capillary.

Research

25 Plasma and other components of blood can survive being separated by centrifuging. Briefly describe how this process is carried out.

Multiple choice quiz

Choose the best response for each question.



26 Which statement about the heart is *not* true?

- **A** It is a specialised muscle.
- **B** It consists of four major chambers.
- **C** It has valves.
- **D** It can be consciously controlled by our brain.

27 Which is *not* a blood vessel?

- **A** vein
- **B** artery
- **C** lymph
- **D** capillary

28 Which of the following is used to listen to a heart beat?

- **A** stethoscope
- **B** sphygmanometer
- **C** electrocardiogram
- **D** endoscope

29 Blood in arteries:

- **A** travels away from the heart.
- **B** travels to the heart.
- **C** remains stationary.
- **D** doesn't have a pulse.

30 Blood is prevented from travelling in the 'wrong' direction by:

- **A** the heart beating.
- **B** valves.
- **C** capillaries.
- **D** glands.

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